

NAME: _____

PERIOD: _____

DATE: _____

Average Payoff, Typical Ticket

AP Statistics · Unit 2 · Topic 2.8 · B: 2026-10-30 · E: 2026-11-20

[STAR] TODAY'S PROBLEM

Suppose a club sells 200 raffle tickets for \$2 each. One ticket wins \$100, four tickets win \$25 each, 20 tickets win \$5 each, and the other tickets win nothing. Let X be the buyer's net profit from one randomly chosen ticket: prize minus ticket cost.

Priya: *The average gross payout is \$1.50 per ticket, so buying one is a good deal.*

Mateo: *Most tickets win nothing.*

Ticket net-profit distribution

Prize	Net x	$P(X = x)$
Top prize	98	1/200
Second prize	23	4/200
Small prize	3	20/200
No prize	-2	175/200

FIRST TAKE — before the video (5 min)

Would you buy one ticket if your goal were to make money? Use one number or outcome from the problem.

[BOOK] DISTRIBUTION, EXPECTATION, AND ONE OUTCOME (keep on the page)

A discrete random variable assigns a number to each random outcome. Its probability distribution lists every possible value x with $0 \leq P(X = x) \leq 1$, and the probabilities sum to 1. The expected value $\mu_X = \sum xP(X = x)$ is the **long-run average** over many repetitions, not the result promised on one trial. For a buyer's **net profit**, subtract the ticket cost from every gross prize before judging the deal.

Finish after the video:

DOK 1 (a) State the four possible values of X and identify the most likely value.

DOK 2 (b) Verify that the probabilities form a distribution. Calculate the expected gross payout and the expected net profit μ_X .

DOK 3 (c) Decide whether Priya's numerical statement and Mateo's statement can both be true, and explain why there is no contradiction. Assess Priya's good-deal conclusion using the ticket cost and μ_X . Explain what expected value does and does not predict for one buyer, then decide whether you would buy and defend the criterion behind your choice.

Frame: Both statements can be true because _____, while the large prizes _____.

Frame: I would _____ because $\mu_X =$ _____ means _____ in the long run, not _____ for one ticket.

EXIT — turn this in

One thing the video changed about my first take: _____